Price Taxing a Model

Do simulated action taxes change how AI agents do research?

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Abstract

I tested whether charging an agent a simulated tax for a research action changes its choices and the quality of its work. Across 96 randomized runs, agents facing a higher price for a full benchmark shifted toward cheaper checks. With an itemized tax, the share using a full benchmark early in the task fell from 100% to 20.8%. Their final model choices also scored 5.2 points lower on a 100-point quality scale, with a 95% interval for the loss of 1.2 to 11.4 points. Raising the price without calling it a tax changed behavior too. The experiment did not establish whether the tax label added an effect beyond an equally priced fee. A separate 32-run experiment found too little variation to determine whether writing out the total price changed choices. These results suggest that prices can influence how an agent gathers evidence, but the reduction in detailed checking came with lower quality on these tasks.

I gave agents a credit budget, varied the price of a full benchmark, and let them choose how to spend their credits. The software deducted the charge when an action occurred or tool was used.

Why study action prices?

The idea came from a few courses I took at Cal, labor economics and behavioral finance classes. I wanted to see whether a research agent would reflect the same behaviors people often make when price changes. My experiment is to grade agent behaviors across three ideas: how people switch tasks when costs change [1], how the way a tax is shown can affect decisions [2], and how AI agents manage a limited budget [3].

I tested this in an experiment where I could check the outcome independently. Each agent had to choose one of six predictive models after gathering evidence about them. The question was whether charging more for the detailed test would reduce its use, and what the agent would give up in return. This is a small step toward understanding how prices might affect AI research.

The experiment

I asked each agent to pick the best of six predictive models. All six were versions of a regression model trained on the same 80 examples, but they used different combinations of inputs and patterns. I gave the models arbitrary letter labels, so the agent could run tests to find out how well each one worked. After it submitted its pick, I scored that model on 1,024 examples the agent had never seen.

I wanted to see whether making the more thorough test cost more would push agents toward quick checks, and whether they would end up picking worse models. I used 12 tasks with synthetic data. For each task, I started four fresh agents and randomly assigned them so that one got each price in the table. All four saw the same task and the same order of model labels. Then I repeated this with four new agents. That gave me eight runs per task, two at each price setting, and 96 runs overall. All sessions used gpt-6.1-sol at low reasoning effort.

Each agent started with 24 experimental credits and could choose up to four tests before submitting. Agents were given three options:

- **Run a quick check.** For 3 credits, the agent could test a model on 12 examples and get a rough idea of how well it worked.
- **Run a full benchmark.** This test used 128 different examples and gave the agent a more reliable estimate. Its price depended on the assigned group.
- **Submit a model.** The agent could submit any of the six models for free. Submitting ended the task, and leftover credits earned no reward.

Each row in the table is one of the four price groups. The middle column shows what the agent saw before deciding whether to run a full benchmark. The last column shows how many credits came out of its budget if it ran one. The benchmark itself was the same in every group.

Assigned group	Full benchmark price shown	Credits charged
Baseline	6 credits total	6
Higher price	10 credits total	10
Itemized fee	6 credits +4 (credit fee) = 10 credits total	10
Itemized tax	6 credits + 4-credit tax = 10 credits total	10

Say the agent runs a full benchmark first. The software deducts the charge as soon as the test runs. A baseline agent spends 6 of its 24 credits and has 18 left. An agent in the tax group spends 10 and has 14 left. The tax applies each time that agent runs a full benchmark, leaving it fewer credits to test other models.

The first two rows let me test how agents respond when the benchmark price rises from 6 to 10 credits. The last three rows all charge 10 credits, so I can compare showing a single price with listing the extra charge separately. Agents in the fee and tax groups saw the same 6-credit base price and the same 4-credit add-on. The benchmark cost 10 credits in both groups. I called the extra 4 credits a fee in one group and a tax in the other. I compared the groups to see if that change in wording affected how often agents ran a full benchmark early or how well they picked a model [2].

Agents ran tests through their normal shell tools. The software enforced the budget and the four-test limit, blocked repeats of the same test on the same model, and kept credit balances and final grading data outside the agents' working directories. I told the agents they were spending experimental credits and that their choices and final results would be recorded and evaluated.

What counts as better outcomes

I wanted to know both whether agents changed their research strategy and whether that change helped them choose a better model. I chose two main measures before collecting the main sample.

1. **Early benchmarking:** Did the agent complete a full benchmark within its first three accepted lab actions? Status requests and rejected requests did not count. Submission ended the run.
2. **Quality of the final choice:** How well did the submitted configuration perform on 1,024 additional examples that the agent never saw?

I looked at the first three actions because agents could still afford a full benchmark during that time. For example, an agent that began with two quick checks had spent 6 of its 24 credits. It still had 18 credits, enough to pay for a full benchmark in any group. Choosing another quick check at that point reflected how the agent spent its budget, while both tools remained available.

I scored the final choices using the unseen examples. Within each task, the best of the six models received 100 points and the worst received 0. A score of 50 meant that prediction error lay halfway between the best and worst results. This gave each task equal weight in the comparison. I also report root mean square error (RMSE), where a lower number means more accurate predictions. I used three comparisons to distinguish a response to price from a response to the tax label.

Tax versus baseline compares agents paying 6 credits for a full benchmark with agents paying 6 credits plus a 4-credit tax. The software deducted 10 experimental credits each time a taxed agent ran a full benchmark. The tax therefore applied to that research action and reduced the agent's remaining budget.

Higher price versus baseline tests what happens when the full benchmark simply costs more.

Tax versus fee changes the label while holding the total and itemization fixed.

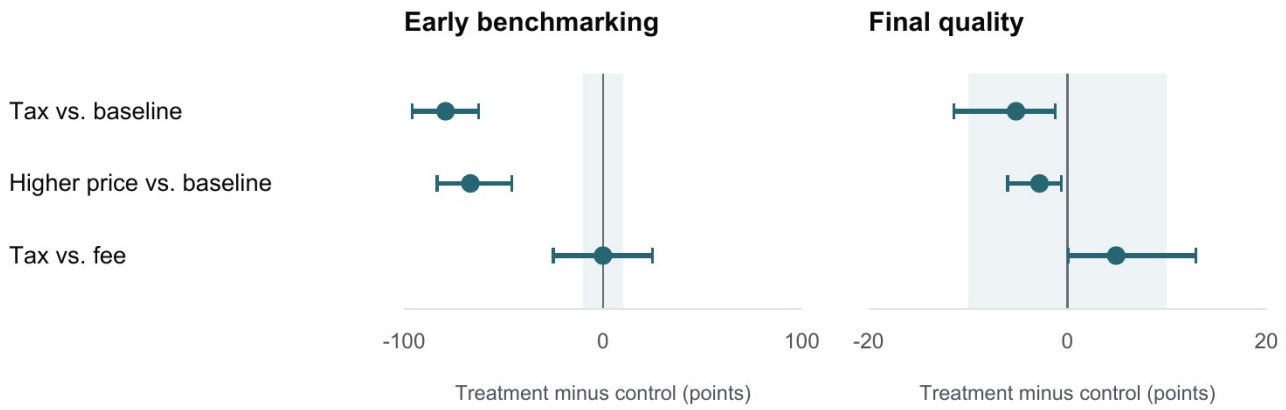
I compared runs within randomized blocks. For the six planned comparisons, covering three contrasts and two main outcomes, I used two-sided paired randomization tests with 19,999 draws and Holm correction. That correction accounts for testing several claims together. The 95% intervals came from resampling whole blocks and describe variation in this fixed task set. I selected a 10-point difference as a practical reference for each main measure. The intervals themselves were not adjusted for multiple comparisons.

How agents changed their choices

All 24 baseline agents ran a full benchmark within their first three accepted actions. That fell to 8 of 24 with a plain higher price and 5 of 24 in each itemized group. The tax-versus-baseline difference was 79.2 percentage points. The plain price increase reduced early benchmarking by 66.7 points. Both results remained statistically significant after accounting for the six planned comparisons. The higher charge shifted agents toward cheaper checks early in the task. This shows that pricing an action can change which evidence an agent chooses to gather.

Group	Early benchmark	Full tests	Quick checks	Quality score	RMSE	Best choice
Baseline	100.0%	4.00	0.00	95.4	0.758	62.5%
Bundled increase	33.3%	1.04	2.88	92.6	0.778	50.0%
Itemized fee	20.8%	0.92	3.00	85.4	0.843	29.2%
Itemized tax	20.8%	1.00	2.96	90.3	0.795	41.7%

24 runs per group. Full tests and quick checks are averages per run. Choosing a configuration uniformly at random would yield an average quality score of 50.6 and RMSE of 1.130 on these tasks.



Dots show estimated differences between groups, and lines show 95% intervals. Early benchmarking is measured in percentage points. Final quality uses the 100-point scale. The shading marks the preselected 10-point reference in either direction. These descriptive intervals are not adjusted for multiple comparisons.

The change was visible from the first decision. All baseline agents benchmarked first. In the tax group, 23 of 24 began with a quick check, although a full benchmark was affordable. The charge therefore influenced the first choice while both tools were available.

Across the whole run, tax agents averaged one full benchmark and almost three quick checks. Baseline agents used all four of their validation calls for full benchmarks. The charge redirected effort toward less precise evidence. Agents continued working, but they relied less on the detailed tool that carried the tax.

The quality cost

Comparison	Early benchmark, points [95% interval]	Adjusted p	Final quality, points [95% interval]	Adjusted p
Tax vs. baseline	-79.2 [-95.8, -62.5]	<0.001	-5.2 [-11.4, -1.2]	0.032
Higher price vs. baseline	-66.7 [-83.3, -45.8]	<0.001	-2.8 [-6.0, -0.6]	0.087
Tax vs. fee	0.0 [-25.0, +25.0]	1.000	+4.9 [+0.1, +12.9]	0.256

The tax group scored 5.2 points lower than the baseline group. Its average prediction error was also higher, at 0.795 compared with 0.758. Ten of the 24 tax agents chose the best configuration, compared with 15 of the 24 baseline agents.

The uncertainty interval for the quality loss ranged from 1.2 to 11.4 points. The estimate remained negative when I removed each task in turn, ranging from a loss of 2.1 to 5.6 points. One task accounted for about 63% of the overall gap, so the size of the loss is sensitive to the task mix.

The plain price increase also had a lower average quality score, but that comparison did not meet the threshold after adjustment for multiple tests. The tax-versus-fee quality comparison was inconclusive. Its descriptive interval narrowly excluded zero, while the planned randomization test did not establish a difference. The same-price label comparison therefore leaves meaningful effects unresolved.

The reduction in detailed checking was substantial. Across all calls, the average number of examples evaluated fell from 512 per run in the baseline group to 163.5 in the tax group, about 68% fewer. Because the lab had precomputed the test results, they do not tell me how much hardware use, inference spending, or time an operator would save. Several limits matter when interpreting the tradeoff:

- The sample covers one hosted model at one reasoning setting and 12 fixed tasks. Repeated sessions improve precision within that set, while generalization to other tasks and models remains open.
- The agent selected among existing configurations. Open-ended research can involve creating methods, running long experiments, and finding different substitutes [4].
- Every agent received research disclosure. I did not separately measure its understanding of that disclosure or compare awareness conditions.
- The outcomes measure checking behavior and final model quality. They provide a starting point for studying resource choices in research.

What price presentation adds

An agent might react to a higher charge, to the way the charge is presented, or to both. The main experiment gives evidence about the first question. To study the second, the amount paid needs to stay fixed.

Chetty, Looney, and Kroft studied the second question with grocery shoppers. They displayed prices including tax without changing what shoppers actually paid. Demand for the affected products fell by about 8%. Their finding motivates testing price presentation separately from a change in the amount charged [2, pp. 1145-1147].

In my main experiment, the higher-price, fee, and tax groups all saw a total of 10 credits. Those comparisons tested labels and itemization when the total was already visible. In the second experiment, I kept the 10-credit charge and the word tax in both groups. I changed whether the total was written out or left for the agent to add up.

This distinction matters for anyone setting an agent's budget. A higher price can limit use of a tool, while clearer price information could change choices without increasing the charge. The results need to show which approach helped.

- If accurate model selection is the priority, the baseline performed better than the itemized tax on these tasks.
- If reducing diagnostic work is the priority, the operator needs to decide how much quality it can afford to lose.
- If the goal is to change behavior through clearer price information, the display experiment needs to establish that effect.

These questions explain why I measured both tool use and final quality. Using fewer full benchmarks could be worthwhile if a cheaper approach still meets a quality target. Here, the tax group gave up some quality. Actual compute savings would need to be measured in a working research system.

Does showing the total matter?

For the second experiment, I held the benchmark price at 10 credits. One group saw ‘6 credits plus a 4-credit tax.’ The other also saw the total written out as 10 credits. Both groups knew that the charge included a tax. I used eight different tasks and 16 fresh sessions per display, for 32 runs in all. The model, reasoning setting, budget, and lab rules stayed the same.

Display	Exact price statement	Charged
Components	6 credits plus a 4-credit tax per benchmark	10
Explicit total	10 credits total (6 credits plus a 4-credit tax)	10

Display	Early benchmark	Full tests	Quality score	RMSE
Components	0.0%	0.88	84.8	0.784
Explicit total	6.2%	0.94	81.1	0.799

16 runs per display. Both conditions charged the same total for a full benchmark. Quality uses the same 100-point scale as the main study.

Early benchmarking occurred in 0 of 16 runs when only the components were shown and 1 of 16 when the total was explicit. The difference was 6.25 percentage points. Mean quality was 84.8 and 81.1 points, respectively, a gap of 3.7 points in the other direction. Both planned randomization tests returned an adjusted p-value of 1.000. The tests found no statistically detectable effect.

- Fifteen of the sixteen paired runs had identical results on both main measures. The remaining pair accounted for both average differences.
- Early benchmarking was close to zero in both groups, leaving little variation to distinguish the displays.
- The comparison was inconclusive. It does not establish that the two displays are equivalent.

Agents could work out the total in either condition. A difference could arise through arithmetic convenience, attention, or wording. The experiment identifies the effect of this particular display change, while its psychological mechanism remains open. The two main tests were specified before collection and adjusted using Holm correction. These results use a separate task set and do not enter the main estimates.

Conclusion and TLDR

The tax changed how agents spent their research budgets. They replaced most full benchmarks with quick checks, and their final model choices were less accurate. For this task, reducing use of the expensive tool came with a quality cost.

- **The behavior changed before the balance forced it.** Agents could still afford a full benchmark when they shifted toward quick checks. The tax group averaged one full benchmark per run, compared with four in the baseline.
- **Less detailed checking came with lower quality.** The tax group's final selections scored 5.2 points lower on the 100-point scale. The estimated loss was uncertain in size and varied with the task mix.
- **The price response is clearer than the label response.** A plain increase also changed early benchmarking. Comparing the same charge as a tax or fee left the extra effect of the word tax unresolved.
- **The display experiment remains inconclusive.** Showing the total explicitly produced too little variation across the 32 runs to establish an effect or equivalence.

These results demonstrate that prices can change which information an agent chooses to obtain. Whether that improves the system depends on what the agent gives up. Reducing an expensive action is valuable when the remaining strategy can still meet the operator's quality requirement.

The baseline produced better model selections throughout the experiment. However, to decide whether the tax is worthwhile in practice, I would need a value for the work saved and a minimum acceptable quality level. Tests across other models, reasoning settings, and research tasks would help establish how widely this result applies at scale.

My next study and experiment will test whether implementing a tax discourages agents from cheating.

Data and references

Every assigned run in both studies completed and submitted a selection. I checked each recorded tool request against the lab's own records, then replayed the credit deductions and final grades.

- The main study contained 96 runs and 571 lab requests. All requests matched the recorded shell commands, with no commands outside the instructed menu and no rejected requests.
- The price-display study contained 32 runs and 192 lab requests. The same checks found no menu violations or rejected requests.
- The frozen study and analysis files matched their recorded hashes after collection. Recomputing the results from the delivered traces reproduced the estimates.

For the replication package, which contains instructions, code, and analysis plans for running both experiments and analyzing new runs, contact me at henrydoan25 398@gmail.com.

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